Shared arithmetic certificates for Diophantine consistency equations

A smaller direct quartic and a degree reduction for the thirteen-integer construction

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Abstract

We construct a fixed quartic polynomial in 667 nonnegative integer unknowns, supplied both as 535 quadratic residuals and as a fully collected list of 23,747 monomials. Its maximum absolute coefficient is 2,649,643,818,040. Existence of a zero is equivalent to existence of a finite ZFC refutation. The construction encodes the numerical proof calculus C_{222} directly, using neither a K215 execution tableau nor Jones's universal polynomial. Relative to the incoming 3,614-variable direct quartic, the reduction gives every binomial test one common radix, all positive-power tests a common Pell certificate, and equal exponents shared Pell coordinates. Exact nonnegative polynomial substitutions then eliminate unnecessary coordinates.

The substantive arithmetic step is a simultaneous-completeness theorem. A single dominating parameter bounds all proposed power outputs; a finite matrix period supplies one Pell solution divisible by the square of the product of all required ordinates. Thus independent local certificates can be replaced by compatible shared witnesses. An unrestricted-integer quartic is also supplied. A separate thirteen-integer polynomial has degree 1072, down from 1088, at the same 195 arithmetic operations. All new equivalences have ordinary proofs formalizable in ZFC. The archive contains the literal equations, builders, exact algebra certificates, verification receipts and source dependencies. No global minimum or new end-to-end proof-assistant certificate is claimed.

Contents

1 The result and its exact interpretation **3**
1.1 What has improved . . . 3
1.2 An explicit polynomial . . . 4

2 The compression principle **4**

3 One radix for every binomial coefficient **5**

4 One offset for signed congruence quotients **5**

5 One Pell certificate for all positive powers **5**
5.1 The two established interfaces . . . 6
5.2 The batched polynomial conditions . . . 6
5.3 Compatibility with a common binomial radix . . . 8
5.4 Eliminating terminal quadratic coordinates . . . 9

6 The direct proof-calculus source **9**
6.1 Finite flow certificates . . . 9
6.2 A carrier recovering the exact flow . . . 9
6.3 Thirteen local coordinates . . . 10

7 Removing the bounded universal quantifier **10**
7.1 Eliminating the remaining arithmetic macros . . . 11
7.2 The positivity order at arbitrary zeros . . . 12

8 The literal quartic and its integer companion **12**
8.1 A safe unrestricted-integer translation . . . 12

9 A secondary reduction: thirteen integers at degree 1072 **13**
9.1 Moving the nonzero scale inside the guard . . . 13
9.2 Compatibility with the five-radical equation . . . 13
9.3 Exact degree, circuit size, and evidence . . . 14

10 Verification and the logical boundary **15**
10.1 Inherited mathematical interfaces . . . 15
10.2 Formalizability and reproducibility . . . 16

11 What this changes about the search **16**

A The fixed numerical calculus **16**

1 The result and its exact interpretation

Let $\mathrm{Ref}_{\mathrm{ZFC}}$ mean that there exists a finite derivation of contradiction from the fixed syntactic ZFC axiom predicate. Thus $\mathrm{Con}(\mathrm{ZFC}) = \neg\,\mathrm{Ref}_{\mathrm{ZFC}}$, with the usual coding of finite proofs. Our goal is a fixed explicit polynomial P for which

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in D^n \; P(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The domain D is part of the assertion. Throughout this report $\mathbb{N} = \{0, 1, 2, \ldots\}$. Strictly positive variables, natural variables and unrestricted integers are not interchangeable without a proved translation.

Theorem 1.1 (Delivered equations)**.** *The literal polynomials specified in this archive satisfy*

$$\begin{aligned}
&\mathrm{ZFC} \vdash (\exists \mathbf{x} \in \mathbb{N}^{667} \; P_{\mathrm{batch}}(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}},\\
&\mathrm{ZFC} \vdash (\exists \mathbf{z} \in \mathbb{Z}^{1690} \; P_{\mathrm{integer}}(\mathbf{z}) = 0) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}},\\
&\mathrm{ZFC} \vdash (\exists \mathbf{u} \in \mathbb{Z}^{13} \; Q_{1072}(\mathbf{u}) = 0) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}\,.
\end{aligned}$$

The two direct polynomials have exact degree four. The thirteen-integer polynomial has exact degree 1072*. Negating these equivalences gives the requested consistency formulations.*

This is an ordinary mathematical theorem obtained from the arguments below and the explicitly identified inherited calculus and arithmetic theorems. It assumes neither consistency nor arithmetic soundness. The unbounded mathematical arguments are separate from the finite software checks. A complete new Lean or object-language ZFC derivation of the emitted coefficient dictionary has not been produced.

1.1 What has improved

The current repository already supplies nine-positive-variable and ten-integer equations in its high-degree families, as well as a thirty-seven-integer quartic [2, 3]. Our direct construction addresses a different useful comparison: degree four, moderate explicit coefficients, and a proof-calculus interpretation without the native machine tableau. It is not a variable-count improvement over those thirty-seven or ten-variable equations.

The direct baseline is the newest incoming `ConZFC_Shared_Certificates_2026-10-24.zip`, rather than the older 16,036-natural-variable direct compiler. The incoming archive contains a completed $3,614$-natural-variable quartic, $2,954$ quadratic rows, $15,996$ monomials and coefficient height $127,778,545$. Its global inventory is 55 power graphs, 16 binomial graphs and two factorial graphs [4].

Natural quartic	Variables	Rows	Monomials	Height
Incoming direct	3,614	2,954	15,996	127,778,545
Common radix	2,108	1,745	10,629	127,777,275
Plus shared offset	1,968	1,717	10,545	127,777,275
Shared Pell frame	1,153	1,021	9,234	127,778,299
Compact natural	667	535	23,747	2,649,643,818,040
Fewer monomials	680	548	18,302	690,871,812,080

Table 1: Exact measurements for the direct natural-variable construction. Every polynomial has degree four. The final integer companion has 1690 variables and 842 quadratic rows.

Height means the maximum absolute coefficient of the expanded polynomial. The support cap in the elimination pass exposes a tradeoff: removing more coordinates can increase the number of monomials and their coefficients. The low-height checkpoint is retained for that reason. Every listed new point is a completed fixed polynomial; no count describes only a fragment of an unfinished compiler.

The second improvement keeps the incoming thirteen-integer norm architecture and its whole fixed coefficient tuple. Moving a square multiplier inside one positivity certificate lowers its exact degree from 1088 to 1072 at unchanged arithmetic cost.

The baseline implementation allocates its Pell environment separately at every power call. The shared-power theorem gives a concrete improvement over that inspected implementation. No worldwide priority claim is made for sharing Diophantine witnesses.

1.2 An explicit polynomial

The primary natural equation is defined by the complete residual lists in `direct/output/compact667/equation.json`. If those fixed lists are $\mathcal{L}_0, \dots, \mathcal{L}_{535-1}$, the equation is exactly

$$P_{\mathrm{batch}}(X_0, \dots, X_{667-1}) = \sum_{\ell=0}^{535-1} \left(\sum_{(c,\mu)\in\mathcal{L}_\ell} c \prod_{j\in\mu} X_j \right)^2 = 0. \tag{1}$$

Each c is a supplied integer and each μ is a sorted list of at most two coordinate indices. The empty list is the constant monomial; a repeated index denotes a square. The variable list fixes the ordering.

This finite notation specifies an ordinary polynomial with fixed integer coefficients. Its collected coefficient dictionary is also included as `direct/output/compact667/expanded_polynomial.json`. No variable exponent, factorial, binomial, remainder, bounded quantifier or unnamed numerical parameter survives. The builder reproduces data already delivered, rather than being the only specification of an as-yet-uncomputed polynomial.

The integer equation has the same two representations in `integer`. The thirteen-integer equation is a division-free straight-line polynomial circuit in `guard13`; its fixed leaves and inherited coefficient provenance are included.

2 The compression principle

An existential arithmetic certificate often chooses auxiliary numbers only to make them sufficiently large, divisible or congruent. Compiling each occurrence separately repeats those choices. We ask whether a single choice can serve an entire finite computation.

Pointwise existence does not by itself justify this: $\forall i\, \exists a\, R_i(a)$ does not imply $\exists a\, \forall i\, R_i(a)$. We prove simultaneous completeness for the particular certificates here. Every binomial digit can use one sufficiently large radix. Every power graph can use one dominating Pell parameter. A finite matrix period meets all of the Pell-ordinate divisibility requirements at once.

The dominating parameter includes the proposed power outputs:

$$W = 1 + \sum_i (n_i + k_i + m_i).$$

At an arbitrary polynomial zero, the global Pell norm forces $A > W$, and the local modulus $T_i = 2An_i - n_i^2 - 1$ then exceeds m_i. Consequently the separate strict output-bound certificates disappear.

There is no self-referential bound. The semantic arithmetic values are chosen before the Pell witnesses. The new Pell coordinates and the common congruence offset do not occur in W. Numerically huge witnesses are compatible with a small fixed number of quantified coordinates.

3 One radix for every binomial coefficient

For finitely many binomial requests $b_j = \binom{a_j}{r_j}$, put $H_0 = \sum_j a_j$ and

$$Q = 2^{H_0} + 1, \qquad L_j = Q^{r_j}, \qquad C_j = (Q+1)^{a_j}.$$

Every actual upper and lower argument is positive; in particular $H_0 > 0$. Since $a_j \le H_0$, every coefficient in the binomial expansion of C_j is at most $2^{a_j} < Q$. Its base-Q digit in position r_j is therefore exactly $\binom{a_j}{r_j}$, including zero if $r_j > a_j$.

Four natural witnesses h_j, u_j, v_j, w_j impose that digit:

$$C_j = L_j(Qh_j + b_j) + u_j, \qquad L_j = u_j + v_j + 1, \qquad Q = b_j + w_j + 1. \tag{2}$$

The last two equations are the exact remainder and digit bounds. Uniqueness of quotient and remainder proves the equivalence.

The baseline has sixteen binomial graphs, each using three powers. The shared construction uses sixteen numerator powers, four distinct denominator powers, and one radix power. Seven other powers remain unchanged. Thus the total inventory falls from 55 to 28 power requests before their Pell certificates are shared.

Some upper arguments depend on earlier factorial or binomial outputs. The builder allocates those semantic outputs first, records all requests, and only then emits the common radix. For completeness choose the intended arithmetic values first; those determine all upper arguments. For soundness an arbitrary polynomial zero already assigns them values, and the power theorem followed by the digit theorem forces their identities. The radix is an auxiliary certificate parameter, not part of the definition of the binomial function.

4 One offset for signed congruence quotients

The baseline represents $L_i \equiv R_i \pmod{D_i}$ by $L_i + D_i a_i = R_i + D_i b_i$, with two natural quotient coordinates. A fixed finite family needs only one common natural offset b:

$$L_i + D_i a_i = R_i + D_i b \qquad \text{for every } i. \tag{3}$$

The forward implication is immediate. Conversely write $L_i - R_i = D_i t_i$ with signed integer t_i, choose $b \ge \max(0, t_1, \dots, t_s)$, and set $a_i = b - t_i$. If $D_i = 0$, the congruence means $L_i = R_i$ and places no restriction on the offset. No coprimality hypothesis is involved.

The common offset is chosen last and excluded from the power-dominating sum. The construction imposes no simultaneous congruence on the offset: it is simply an upper bound on finitely many signed quotients.

5 One Pell certificate for all positive powers

The principal new compression is at the level of existential certificates. Every requested power retains its own input and output, but the auxiliary number-theoretic environment is shared by the entire finite computation. This is stronger than reusing identical arithmetic subexpressions: it restricts independently chosen old witnesses to a common family, whose simultaneous completeness must be proved.

For an integer $a \ge 2$, let $\chi_a(t), \psi_a(t) \in \mathbb{N}$ be the Pell coordinates defined by

$$\chi_a(t) + \psi_a(t)\sqrt{a^2 - 1} = (a + \sqrt{a^2 - 1})^t.$$

These functions occur only in the proof. The emitted equation contains ordinary polynomial expressions and no Pell-function symbol. Write

$$\mathcal{N}(a, x, y): \quad x^2 + y^2 = 1 + a^2 y^2.$$

All variables in this section range over $\mathbb{N} = \{0, 1, 2, \ldots\}$.

5.1 The two established interfaces

We use the positive-index case of the classical Pell-index representation. For $a \geq 2$ and $k > 0$, the conjunction

$$k \leq Y, \qquad \mathcal{N}(a, X, Y), \qquad \mathcal{N}(a, U, V), \qquad \mathcal{N}(B, S, R), \tag{4}$$

$$B \geq 2, \quad B \equiv 1 \pmod{4Y}, \quad B \equiv a \pmod{U}, \quad V > 0, \quad Y^2 \mid V, \tag{5}$$

$$S \equiv X \pmod{U}, \qquad R \equiv k \pmod{4Y} \tag{6}$$

is equivalent to $X = \chi_a(k)$ and $Y = \psi_a(k)$. In particular the condition $k \leq Y$ makes $Y > 0$ before any congruence modulus is used.

The positive-power representation says that, for $n, k > 0$, $m = n^k$ is equivalent to the existence of W, a, T, Z, X, Y with $a \geq 2$ satisfying

$$X = \chi_a(k), \qquad Y = \psi_a(k), \tag{7}$$

$$X + Yn \equiv Ya + m \pmod{T}, \qquad 2an = T + n^2 + 1, \tag{8}$$

$$m < T, \qquad n \leq W, \qquad k \leq W, \qquad \mathcal{N}(W + 1, a, WZ). \tag{9}$$

These are the interfaces formalized as `Pell.matiyasevic` and `Pell.eq_pow_of_pell` in Mathlib's `PellMatiyasevic.lean` [6]. The shifted congruence in (8) avoids truncated subtraction. The standard proof derives $a \geq n$, so it is equivalent to the source formulation $X \equiv Y(a - n) + m \pmod{T}$.

5.2 The batched polynomial conditions

Consider a fixed nonempty list of power requests

$$(n_i, k_i, m_i), \qquad 1 \leq i \leq s,$$

whose bases and exponents are positive. Group identical exponent expressions into $e_1, \ldots, e_g$, writing $k_i = e_{j(i)}$. Equality of expressions is used here, not agreement at sampled numerical inputs. Let

$$W = 1 + \sum_{i=1}^{s} (n_i + k_i + m_i).$$

Duplicate expressions may be included just once, provided the resulting natural sum remains strictly larger than every base, exponent, and output. Introduce a single frame A, Z, U, V, B, one quadruple (X_j, Y_j, S_j, R_j) for each exponent group, and one local modulus T_i for each power request. Put

$$P = \prod_{j=1}^{g} Y_j.$$

The global conditions are

$$A \geq 2, \quad \mathcal{N}(W + 1, A, WZ), \quad \mathcal{N}(A, U, V), \tag{10}$$

$$V = P^2(d + 1), \qquad B = 1 + 4Pq, \qquad B \equiv A \pmod{U}. \tag{11}$$

Here d, q are natural witnesses. For each exponent group retain

$$e_j \le Y_j, \quad \mathcal{N}(A, X_j, Y_j), \quad \mathcal{N}(B, S_j, R_j), \tag{12}$$

$$S_j \equiv X_j \pmod{U}, \qquad R_j \equiv e_j \pmod{4Y_j}. \tag{13}$$

For each request retain only

$$2An_i = T_i + n_i^2 + 1, \qquad X_{j(i)} + Y_{j(i)} n_i \equiv Y_{j(i)} A + m_i \pmod{T_i}. \tag{14}$$

The separate output-bound, W-bound, B-bound, local V-divisibility, and local B-congruence certificates have disappeared.

Theorem 5.1 (Simultaneous power certificate)**.** *For every finite nonempty list of positive bases and exponents, the conditions (10)–(14) have natural witnesses if and only if $m_i = n_i^{k_i}$ for every i. If at least one base is known to be at least two, the explicit guard $A \ge 2$ may be omitted.*

Soundness. First suppose $A \ge 2$. The last norm excludes $Z = 0$, since that would give $A = 1$. Thus $Z \ge 1$, and

$$A^2 = 1 + (W^2 + 2W)W^2 Z^2 > W^2.$$

Consequently $A > W$, which bounds every input and proposed output. Each $Y_j \ge e_j > 0$, so $P > 0$ and

$$V = P^2(d + 1) \ge Y_j^2 \ge Y_j > 0.$$

In particular $Y_j^2 \mid V$ and $B \equiv 1 \pmod{4Y_j}$.

The omitted guard $B \ge 2$ follows from the retained equations. We have $B = 1 + 4Pq \ge 1$. If $B = 1$, then $\mathcal{N}(B, S_j, R_j)$ gives $S_j = 1$. The two norms with parameter A, together with $V \ge Y_j$, give $U \ge X_j \ge 2$. The congruence $S_j \equiv X_j \pmod{U}$ would then imply $U \mid X_j - 1$, although $0 < X_j - 1 < U$. This is impossible.

All hypotheses of the Pell-index representation now hold group by group, so $X_j = \chi_A(e_j)$ and $Y_j = \psi_A(e_j)$. For the omitted output bound, $A \ge n_i + 1$ yields

$$T_i - A = (2n_i - 1)A - n_i^2 - 1 \ge (n_i + 2)(n_i - 1) \ge 0.$$

Hence $m_i < W < A \le T_i$. The standard positive-power representation applies separately to every request and proves $m_i = n_i^{k_i}$.

Finally suppose the guard $A \ge 2$ was omitted, but some $n_i \ge 2$. The retained modulus definition gives $2An_i \ge n_i^2 + 1 > 2n_i$, and therefore $A > 1$ before any preceding argument is used. In the actual compiler a literal base-two request gives the particularly simple equation $4A = T_i + 5$. □

Simultaneous completeness. Assume $m_i = n_i^{k_i}$ for every request, and first form the finite natural number W from these semantic inputs and outputs. Choose

$$A = \chi_{W+1}(W), \qquad Z = \frac{\psi_{W+1}(W)}{W}.$$

The quotient is an integer because $\psi_{W+1}(t) \equiv t \pmod{W}$. Also $W > 0$, $A \ge 2$, and $\mathcal{N}(W + 1, A, WZ)$ holds. For each i,

$$n_i^{k_i} \le n_i^W < (W + 1)^W \le A.$$

The standard Pell power congruence therefore supplies a positive $T_i = 2An_i - n_i^2 - 1 > m_i$ and the local congruence (14). This is the source proof of the power representation with a common upper bound in place of $\max(n_i, k_i)$.

Set $X_j = \chi_A(e_j)$ and $Y_j = \psi_A(e_j)$, so all $Y_j > 0$, and form $P = \prod_j Y_j$. In the finite ring $\mathbb{Z}/(4P^2)$, the matrix

$$M = \begin{pmatrix} A & A^2 - 1 \\ 1 & A \end{pmatrix}$$

is invertible because its determinant is one. Its powers belong to a finite group. Some positive power is therefore the identity. Taking a sufficiently large multiple L of that period ensures $L > \max_j e_j$ and $L > 1$. Put $U = \chi_A(L)$ and $V = \psi_A(L)$. The Pell matrix formula gives

$$U \equiv 1 \pmod{4P^2}, \qquad V \equiv 0 \pmod{4P^2}.$$

Thus $V > 0$, $P^2 \mid V$, $U > A$, and $\gcd(U, 4P) = 1$. The Chinese remainder theorem gives $B \geq 2$ with

$$B \equiv A \pmod{U}, \qquad B \equiv 1 \pmod{4P}.$$

Indeed the least nonnegative solution is already at least A, since $A < U$. Define $d = V/P^2 - 1$ and $q = (B-1)/(4P)$.

For each group choose $S_j = \chi_B(e_j)$ and $R_j = \psi_B(e_j)$. The recurrence with parameters congruent modulo U yields $S_j \equiv X_j \pmod{U}$. The recurrence modulo $B - 1$ yields $R_j \equiv e_j \pmod{4Y_j}$. All norms and bounds hold by construction, so every retained condition is satisfied simultaneously. □

The finite-group step is deliberately stated for modulus $4P^2$. Separately proving $Y_j^2 \mid V$ for every j would not prove $P^2 \mid V$, because the Y_j can share prime factors. The common period argument supplies the stronger divisibility actually used by the emitted certificate.

5.3 Compatibility with a common binomial radix

The direct compiler also replaces the separate digit-extraction radices of its finitely many binomial graphs by one radix. Given the positive upper arguments at the actual call sites, $a_1, \ldots, a_r \in \mathbb{N}$, choose

$$H = \sum_{\ell=1}^{r} a_\ell, \qquad Q = 2^H + 1.$$

Then $Q > 2^{a_\ell}$ for every ℓ. Every coefficient in $(Q+1)^{a_\ell}$ is smaller than Q, since $\binom{a_\ell}{j} \leq 2^{a_\ell}$. Consequently quotient and remainder equations for the digit at position b_ℓ recover $\binom{a_\ell}{b_\ell}$ exactly, including the value zero when $b_\ell > a_\ell$. Repeated lower arguments share the same power Q^{b_ℓ}. This is an ordinary positional-notation identity, with no probabilistic or approximate digit extraction.

This common-radix construction and Theorem 5.1 are compatible in a definite order. First choose a semantic solution of the parent certificate and the common radix. Its finite power-request list then has definite natural inputs and correct outputs. Next form W including all those outputs and choose the common Pell frame. No Pell-frame coordinate enters the definition of the common radix or the list of semantic power requests. Conversely, any zero of the final equation yields all these natural values at once; the batched theorem forces every requested power to have its intended value before the binomial digit argument is invoked.

Both implications concern all natural numbers, including arbitrarily large witnesses. A search over a finite numerical range is neither used nor offered as a substitute for these proofs. The argument consists of finite arithmetic, elementary Pell facts, finite-group periodicity, and the Chinese remainder theorem, all formalizable in ZFC. Composing it with the parent's direct C222 calibration preserves the stated ZFC-provable consistency equivalence; it does not establish a new kernel-checked or object-language ZFC derivation by itself.

5.4 Eliminating terminal quadratic coordinates

The final simplification acts on the literal quadratic residual system, independently of the Pell implementation. Suppose one residual is

$$v - f(\mathbf{x}) = 0,$$

where $f \in \mathbb{N}[\mathbf{x}]$ has degree at most two and does not contain v. If f is affine, substitute it for v everywhere. If f is quadratic, make the same substitution provided v occurs only linearly in every other residual. Every new residual still has degree at most two. Deleting the defining row and the variable v gives an exact bijection of natural zero sets: projection in one direction and the unique restoration $v = f(\mathbf{x}) \in \mathbb{N}$ in the other. The nonnegative coefficients are a sufficient, explicitly checked condition for preserving the natural domain; a signed linear-algebra elimination would not have this property automatically.

Repeated application provides a compact certificate consisting of the deleted row, variable, and restoration polynomial at each step. Reading the certificate backwards reconstructs every removed coordinate. The verification checks exact polynomial identities, the domain condition, and degree preservation at every step. It is stronger than comparing the final polynomials at a selection of sample inputs.

6 The direct proof-calculus source

This section records the inherited source mechanism [4, 5] so that the consistency interpretation is explicit. The new compression acts on the whole source, not merely a local arithmetic relation.

The fixed numerical calculus C_{222} uses implication code $I(a, b) = 2(a + b)(a + b + 1) + 4b$, with offsets $1, 2, 3$ for membership, universal quantification and equality. Negation is $I(a, 1)$. Appendix A gives all constructors and twelve rule branches. Truth is the freely available code 0; contradiction is code 1. The numerical calculus is calibrated to syntactic ZFC refutability by the named inherited theorems in Section 10.

6.1 Finite flow certificates

A padded rule with premises p, q and conclusion v has vector $e_v - e_p - e_q$, where $e_0 = 0$ and positive labels are independent basis vectors. A finite list of valid instances satisfying

$$\sum_i (e_{v_i} - e_{p_i} - e_{q_i}) = e_1 \tag{15}$$

already implies derivability of 1. Let D be the closure under the listed rules of truth and their axiom conclusions. If $1 \notin D$, sum the coordinates outside D. A conclusion outside D has at least one premise outside D, so every rule contributes at most zero there. The right side contributes one, a contradiction. Only finitely many labels occur, so this is a finite sum. Cycles and repeated premises cause no exception.

Conversely a finite proof tree gives the flow by counting its nodes. Expand it starting from the open obligation 1, replacing each conclusion by its selected premises.

6.2 A carrier recovering the exact flow

Let $t, m \geq 1$ be the number of instances and a bound on their labels, and put $M = (m + 1)^t$. Set

$$A(X) = \prod_{i<t} (1 + v_i X), \qquad C(X) = (1 + X) \prod_{i<t} (1 + p_i X)(1 + q_i X).$$

Both have nonnegative coefficients and constant term one. Then

$$A(M) = C(M) \quad \Longleftrightarrow \quad A(X) = C(X). \tag{16}$$

If $\deg C \geq t+1$, its evaluation is at least M^{t+1}, whereas

$$A(M) \leq (1+mM)^t < ((m+1)M)^t = M^{t+1}.$$

Thus each product has at most t nonconstant factors and coefficient sum at most M. Every individual coefficient is strictly below M, because the constant term is one and a nonconstant product has another positive coefficient. Every coefficient of $A - C$ therefore has absolute value below M. Its least nonzero coefficient, if any, would be divisible by M, by reducing its zero evaluation after removing the corresponding power of M. This is impossible.

Unique factorization in $\mathbb{Q}[X]$ now recovers the exact multiplicities of the distinct factors $1 + jX$ for $j > 0$, giving (15). No coprimality of their integer values is assumed.

Natural accumulator states satisfy

$$S_0 = M + 1, \quad S_t = 1, \quad S_{i+1}(1 + v_i M) = S_i(1 + p_i M)(1 + q_i M). \tag{17}$$

Positive factors force every state positive, and telescoping gives (16). For completeness take S_i to be the product of the weights of current open obligations in the proof-tree expansion. Every update is then an integer identity.

One beta code represents the accumulator: $\beta(x, y, i) = x \bmod (1 + (i+1)y)$. Choose y sufficiently large and divisible by every relevant positive index difference. The moduli are pairwise coprime and exceed all sequence entries, so CRT supplies x. The choice order is finite proof, t, m, carrier, accumulator, then x, y. The later witness bound is independent of the earlier label bound m.

6.3 Thirteen local coordinates

The local row uses three labels p, q, v, four schema coordinates a, b, c, d and six beta/accumulator coordinates S, S', u, r, u', r'. For $D_i = 1 + (i+1)y$ its arithmetic rows are

$$\begin{aligned} x &= S + D_i u, & D_i &= S + r + 1, \\ x &= S' + (D_i + y)u', & D_i + y &= S' + r' + 1, \\ S'(1 + vM) &= S(1 + pM)(1 + qM). \end{aligned}$$

The first four rows give exact consecutive beta digits, since the divisors are positive.

For each rule branch square and sum its residuals; multiply the twelve branch sums to impose their disjunction. Add the squares of the five other local rows. This is a fixed nonnegative polynomial $L(i, \mathbf{z}, \mathbf{w})$. Its implementation uses nonnegative-coefficient circuits E, F with $L = E - F$. For multiplication,

$$(E_1, F_1)(E_2, F_2) = (E_1E_2 + F_1F_2,\ E_1F_2 + F_1E_2).$$

Addition and subtraction use componentwise addition and the appropriate swap. The difference is preserved and both components are monotone in natural inputs.

7 Removing the bounded universal quantifier

Give the local witnesses bounds u_j and choose $V \geq u_j$ for every j. Define

$$H = E(t, \mathbf{z}, \mathbf{u}) + F(t, \mathbf{z}, \mathbf{u}) + t + V + 1, \qquad T = H!, \qquad B_0 = \prod_{i<t}(1 + (i+1)T).$$

Existence of bounded witnesses for every index $i < t$ is equivalent to natural $C, A_1, \ldots, A_d$ satisfying

$$B_0 \mid \binom{A_j}{u_j + 1} \qquad \text{for each } j, \tag{18}$$

$$B_0 \mid 1 + (C+1)T, \tag{19}$$

$$E(C, \mathbf{z}, \mathbf{A}) \equiv F(C, \mathbf{z}, \mathbf{A}) \pmod{B_0}. \tag{20}$$

Soundness. Fix $i < t$ and a prime divisor p of $D_i = 1 + (i+1)T > 1$. It is coprime to $T = H!$, so $p > H$. The integer identity

$$(u_j + 1)!\binom{A_j}{u_j + 1} = \prod_{r=0}^{u_j} (A_j - r)$$

also holds for $A_j \le u_j$, when both sides vanish. Primality supplies a $w_j \in [0, u_j]$ with $A_j \equiv w_j \pmod{p}$. Comparing the index divisibility with $D_i \equiv 0$ and cancelling T gives $C \equiv i \pmod{p}$. The row congruence yields $E(i, \mathbf{z}, \mathbf{w}) \equiv F(i, \mathbf{z}, \mathbf{w}) \pmod{p}$. Monotonicity places both natural values strictly below p, hence they are equal. □

Completeness. A common divisor of D_i, D_j for $i < j < t$ divides $(j - i)T$ and is coprime to T, so it divides $j - i$. But $j - i \mid H!$, making the common divisor one. CRT now chooses $C \equiv i$ and $A_j \equiv w_j^{(i)} \pmod{D_i}$ for valid bounded row witnesses. Their falling-factorial products vanish modulo every D_i. Since $(u_j + 1)! \mid T$ and $\gcd(B_0, T) = 1$, that factorial cancels, giving the binomial divisibility. The index and polynomial congruences follow by substitution and pairwise coprimality. □

Soundness deliberately uses a prime divisor of each modulus. A product vanishing modulo a composite number does not by itself supply a small root. Completeness deliberately establishes $H \ge u_j + 1$ before factorial cancellation.

For the actual source, the three labels have bound m and the other ten local coordinates have bound $V = m + U_0$ for an extra natural U_0. There are thirteen range-binomial graphs. Endpoint equations directly encode $S_0 = M + 1, S_t = 1$. Writing $t = t_0 + 1, m = m_0 + 1$ loses no proof: zero length contradicts the endpoints, and zero label bound makes every update constant. Encoded coordinates may be shifted to $A_j = \alpha_j + 1$, since adding a multiple of B_0 preserves the falling-factorial divisibility and its factorial cancellation.

7.1 Eliminating the remaining arithmetic macros

The explanatory product is replaced by

$$Q_0 = (1 + (t+1)T)^t + 2, \qquad B_0 = \left(T^t\, t!\binom{Q_0 + t}{t}\right) \bmod (TQ_0 - 1).$$

The numerator is $\prod_{i=1}^{t}(TQ_0 + iT)$ and reduces to the intended product modulo $TQ_0 - 1$. That product is smaller than the positive modulus, so it is the exact remainder.

For a natural n, let $\rho = (n+2)^{n+2}$, $C = \binom{\rho+n}{n}$ and $P = (\rho+1)^n$. Then

$$n! = \lceil P/C \rceil .$$

For $n \ge 2$,

$$\frac{P}{C} = n! \prod_{i=1}^{n} \left(1 - \frac{i-1}{\rho + i}\right), \qquad 0 \le n! - \frac{P}{C} < \frac{n!\, n(n-1)}{2(\rho+1)} < 1.$$

The cases $0, 1$ are immediate. Natural r, s impose the ceiling by $P + C = Cf + r + 1$ and $C = r + s + 1$. Each other division uses the analogous quotient/remainder equations. All binomials use the shared radix, and all powers use the shared Pell certificate. No arithmetic macro remains in the final polynomial.

7.2 The positivity order at arbitrary zeros

All twenty-three exponent groups are positive from structural shifts and nonnegative arithmetic, before any power output is known correct. Consequently every group has $Y_j \geq e_j > 0$ and the common index certificate recovers all its Pell pairs first.

Twenty-seven of the twenty-eight power bases are structurally positive. The remaining base is the factorial output T, which must be proved positive rather than silently assumed. Apply the local power argument first to the structurally positive bases. In particular the factorial numerator $P = (\rho + 1)^H$ is positive. The retained ceiling rows $P + C = CT + r + 1$ and $C = r + s + 1$ would imply $P \leq 0$ if $T = 0$. Hence $T > 0$ and the final power T^t is valid.

The common radix power has literal base two and a positive exponent, so its correctness is available before the binomial digit argument. Its coefficient bound depends only on the assigned natural upper arguments, not on their prior semantic correctness. This order resolves the apparent cycle introduced by deferring factorial and binomial outputs.

8 The literal quartic and its integer companion

Intermediate expressions use natural inputs, nonnegative numeral leaves, addition and multiplication. Each signed identity is stored as equality of two such expressions. For a distinct arithmetic node introduce z and impose $z - x - y = 0$ or $z - xy = 0$. The gate graph is ordered, so its values restore uniquely from semantic witnesses. Squaring and summing its residuals produces one quartic.

The arithmetic gate graph remains acyclic when semantic power outputs are allocated before their certificates: those outputs are existential inputs of the graph, not undefined gate values. Their intended meanings are proved by the simultaneous theorem.

The inherited affine pass removes $z = L$ when L is affine with nonnegative coefficients. The new pass also removes $z = F$ for a nonnegative quadratic F if z occurs only linearly in every other remaining row. Every deletion has a unique natural restoration and preserves degree at most two in the residuals. The complete certificate records every row, coordinate and restoration polynomial. Its reverse order restores all deleted coordinates.

8.1 A safe unrestricted-integer translation

For any natural coordinate x, the integer equation

$$x = a^2 + b^2 + c(c + 1) \tag{21}$$

is an exact nonnegativity guard. Its right side is nonnegative for all integers a, b, c. Conversely the three-square theorem represents $4x + 1$; modulo four exactly one of the three squares is odd and the other two are even. Dividing by four gives (21).

We guard the semantic coordinates of the shared-frame checkpoint *before* the final natural-coordinate elimination. Its retained gate coordinates are then nonnegative by topological induction from their nonnegative arithmetic definitions. The integer builder checks this entire dependency order. Guarding only the coordinates named as witnesses after a later elimination would require a new sign proof, because a restored witness can depend on retained gate values. We do not assume that inference.

After the guards have been added, monic substitutions over $\mathbb{Z}$ can reduce the augmented quadratic system. They preserve its exact integer zero set and all remaining residual degrees. Their restoration polynomials need not preserve positivity, since that condition is already expressed by retained guard equations and their substitutions. The delivered integer polynomial has 1690 coordinates, 842 quadratic rows and exact degree four; it is supplied as residuals and as collected coefficients.

9 A secondary reduction: thirteen integers at degree 1072

This section concerns the separate, large-coefficient Pell construction in the incoming `ConZFC_13_Integer_Norms_2026-10-24.zip`. It is independent of the direct compiler developed above. The parent CRT construction has thirteen unrestricted integer variables, exact degree 1088, and 195 live binary arithmetic operations. One change to its joint positivity guard lowers the degree to 1072, with the same thirteen variables, the same 195 operations, and exactly the same six fixed integer coefficients. This is a degree-only improvement.

9.1 Moving the nonzero scale inside the guard

For integers x, y, t put

$$\mathrm{Pair}(x,y,t) = xy - (t^2 - x - y + 1)^2, \qquad \mathrm{Pos}(p_1,p_2,p_3) = 1 + p_1^2 + p_2^2 + p_3(p_3+1).$$

The two elementary interfaces are

$$\begin{aligned} x > 0 \ \wedge \ y > 0 \quad &\Longleftrightarrow \quad \exists t \in \mathbb{Z} \ \ \mathrm{Pair}(x,y,t) > 0, \\ n > 0 \quad &\Longleftrightarrow \quad \exists p_1,p_2,p_3 \in \mathbb{Z} \ \ n = \mathrm{Pos}(p_1,p_2,p_3). \end{aligned}$$

For the first, positivity implies $xy > 0$; two negative arguments make the subtracted square exceed xy. For positive arguments, a nearest square to $N = x + y - 1$ has squared error strictly less than $N \le xy$. The second follows from the three-square theorem applied to $4n - 3$; its residue modulo four gives one odd square and two even squares.

It follows that both equations

$$v^2 \,\mathrm{Pair}(G,L,t) = \mathrm{Pos}(p_1,p_2,p_3), \tag{22}$$

$$\mathrm{Pair}(v^2 G, L, t) = \mathrm{Pos}(p_1,p_2,p_3) \tag{23}$$

express exactly $v \neq 0$, $G > 0$, and $L > 0$ after existential quantification of the four private guard variables. Their private witnesses need not be the same. The benefit is algebraic: when L has substantially higher degree than $v^2 G$, the leading L^2 term in (23) avoids the outer factor v^2.

9.2 Compatibility with the five-radical equation

Retain the parent's thirteen coordinates

$$j, C, z, h, r, v, w, T, c, n, t, p_2, p_3 \in \mathbb{Z}$$

and every native, transport, Pell, and carrier definition. In particular,

$$J = 6j + 1, \quad q = \beta J + 1, \quad W = 12w + 4, \quad X = qW, \quad Y = qv, \quad a = (X+1)Y,$$

$$H = 4a + 3, \quad \Delta = a^2 + H, \quad \eta = c - Yk, \quad \zeta = k - \eta,$$

and

$$G = (A_{\mathrm{cone}}F)(B_{\mathrm{cone}}C) - (q - A_{\mathrm{cone}}F - B_{\mathrm{cone}}C)^2.$$

Replace only the guard radicand by

$$A_{\mathrm{g}}^{\mathrm{new}} = v^2 G\eta\zeta - (t^2 - v^2 G - \eta\zeta + 1)^2 - 1 - p_2^2 - p_3(p_3+1). \tag{24}$$

The single polynomial is the same cleared five-radical norm,

$$\delta^{16}\mathcal{F}_{32}\Big(Z; M^2\Lambda^2 D_{\mathrm{m}}, U^2 D_f, U^2 D_\tau, U^2 A_{\mathrm{g}}^{\mathrm{new}}, U^2 C_y/\delta\Big), \tag{25}$$

where $M = 2q^2$, $\Lambda = 4H+1$, and $U = MH$. Here $\mathcal{F}_{32}$ denotes the product over all thirty-two independent signs. The supplied arithmetic graph clears the denominator and eliminates the formal radicals; its operations are only $+, -, \times$. The complete front end, coefficients, and literal graph accompany this supplement, together with the complete unchanged parent package.

The parent's unconditional exceptional-case proof still gives $U \neq 0$, $\delta \neq 0$, and four strictly positive non-guard radicands at every integer assignment, including $v = 0$ and $c = 0$. At a norm zero, a negative $A_{\mathrm{g}}^{\mathrm{new}}$ would therefore contribute the only nonzero imaginary summand of a vanishing conjugate factor. Thus $A_{\mathrm{g}}^{\mathrm{new}} \geq 0$ and

$$\mathrm{Pair}(v^2 G, \eta\zeta, t) = A_{\mathrm{g}}^{\mathrm{new}} + 1 + p_2^2 + p_3(p_3+1) > 0.$$

The guard lemma recovers $v \neq 0$, $G > 0$, and $\eta\zeta > 0$ in the required order. In particular $c = 0$ is impossible, because it would give $\eta\zeta = -Y(Y+1)k^2 \leq 0$.

The separation bound is unchanged. The parent proves $G < q^2$, while $\eta\zeta \leq k^2/4$, so

$$\sqrt{A_{\mathrm{g}}^{\mathrm{new}}} < |v|\sqrt{G\eta\zeta} < \frac{|v|\,|q|\,|k|}{2}.$$

This is exactly its former bound for the guard radical. Consequently the same strict ordering of radical magnitudes, rational-square extraction, Pell descent, and two coprime congruences restore the same calibrated source kernel.

For completeness start from a canonical source witness. Since $v^2 G$ and $\eta\zeta$ are positive, choose a new t by the guard lemma and then a new positive representation with coordinates p_1, p_2, p_3. This makes (24) equal to p_1^2. All other roots and source coordinates retain their values. The same integral choice of center,

$$n = \Lambda\gamma + f + \tau + p_1 + y - V - (v - q^2)/M,$$

makes one norm factor vanish. Thus the new and old equations are existence-equivalent over unrestricted integers. No identification of their arbitrary private guard assignments is asserted.

9.3 Exact degree, circuit size, and evidence

The degree-seven leading part η_0 of η gives the new degree-twenty-eight guard component $-\eta_0^4$, replacing the old degree-thirty component $-v^2\eta_0^4$. Hence the scaled guard radicand $e = U^2 A_{\mathrm{g}}^{\mathrm{new}}$ has degree forty rather than forty-two. The independently expanded biquadratic cut has component degree rows

$$(62, 58, 67, 53), \qquad (134, 120, 129, 125), \qquad (268, 254, 263, 259).$$

Its unique highest component is scalar. The final two quadratic norms square it twice, giving exact total degree $4 \cdot 268 = 1072$. An explicit coefficient on the actual fixed-coefficient slice is

$$[j^{480} w^{160} v^{192} h^{32} r^{64} c^{128} z^{16}]P = 2^{976} 3^{640} \beta^{480} > 0.$$

The exponent sum is 1072, establishing attainment of the degree bound independently of the other coefficient values.

The former multiplication $v^2\,\mathrm{Pair}(G, L, t)$ is replaced by $v^2 G$. The literal evaluator still has 41 additions, 122 multiplications, and 32 subtractions, totaling 195. Normal and optimized Python verification authenticate the whole inherited source manifest, fixed numeral payloads, exact cut identities, leading coefficient, and literal graph regeneration. Modular evaluations against a separate conjugate-product evaluator are labeled transcription diagnostics, rather than universal proofs.

The new theorem is this ordinary arithmetic existence reduction. Its Con(ZFC) interpretation inherits the parent's named native/Pell and K215 calibration arguments. The supplement supplies neither a new end-to-end proof-assistant certificate nor a new object-language ZFC derivation, and does not assert that the polynomial has no zero.

10 Verification and the logical boundary

The verification has three roles. Ordinary mathematics proves the unbounded existence equivalences. Symbolic audits bind the implemented source and arithmetic graphs to those arguments. Exact coefficient collection and certificate replay bind the final polynomial to the graph. No finite check decides whether the consistency equation itself has a zero.

The archive checks the fixed numerical calculus, all shared-radix requests, every positive-power call site, exact exponent grouping, the complete dominating sum, the product of ordinates, the common norms and congruences, every local modulus, and the exclusion of certificate coordinates from the dominating sum. It replays each elimination as an integer-polynomial identity and independently expands every delivered squared sum. Degree, coefficient height, coordinate closure and variable use are measured from the actual data.

Fixtures supplement those identities. They exercise signed congruence quotients, binomial positions beyond the degree, shared Pell periods and false power outputs. They are small arithmetic examples, not purported ZFC refutations. The receipts distinguish normal and optimized Python runs. Required checks raise explicit errors rather than depending on assertions disabled by optimization.

10.1 Inherited mathematical interfaces

The direct source uses the calibrated equivalence

$$C_{222}(1) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZF}} \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}\,.$$

The named repository declarations are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

They concern the fixed syntactic predicates. The second uses constructible-universe equiconsistency; neither assumes consistency. Relevant source copies, notices and pins accompany the direct component.

The classical Pell-index and positive-power representations are named interfaces in Mathlib and the repository's literal representation modules [6]. This contribution proves that their witnesses can be selected simultaneously. It does not reprove the full classical index theorem.

The thirteen-integer supplement has a separate inherited native/Pell calibration. Packaging it beside the direct quartic does not remove or replace that dependency.

10.2 Formalizability and reproducibility

All new arguments use finite lists, polynomial identities, elementary inequalities, Pell recurrences, finite groups, finite CRT and finite proof transformations. The matrix-period step is a pigeonhole argument in a finite group. These arguments are available in ZFC for its internal natural numbers, with no restriction to standard witnesses. Composing them with the identified calibrations proves Theorem 1.1.

This is not a proof of Con(ZFC) or a new end-to-end formal-kernel certificate. Under the usual consistency hypothesis, the no-zero assertion is unprovable in ZFC by its equivalence with consistency and the second incompleteness theorem. No claim of two-sided independence from mere consistency alone is needed.

The archive's `README.md` gives exact reproduction commands. Its top-level `verify.py` checks the delivered data and rebuilds selected artifacts in temporary directories. The TeX source compiles with `pdflatex -interaction=nonstopmode -halt-on-error report.tex`, run twice for references. The contribution is prepared for incorporation in ProveIt; it makes no remote repository change.

11 What this changes about the search

The direct compiler's repeated cost was largely the existential infrastructure for arithmetic functions. Optimizing that infrastructure as a finite family makes a substantial reduction possible before any generic flattening. The simultaneous-completeness proof is the reusable result: sharing is justified by common size bounds and finite periods, rather than assumed from separate existential statements.

Further work could exploit proved relations between different exponents, establish more forced signs in the integer system, or translate the finite certificates into a proof-assistant development. None of those prospective gains is included in the present counts.

The thirteen-integer equation retains very few variables with large inherited constants. The direct equation has degree four, moderate fixed coefficients and a proof-calculus interpretation. These are different useful improvements; no comparison here proves a global simplest Diophantine equation.

A The fixed numerical calculus

The following inherited numerical source [4] makes the fixed logical content of the coefficient dictionary explicit. For natural inputs define

$$I(a,b) = 2(a+b)(a+b+1) + 4b,$$
$$M(a,b) = I(a,b) + 1, \qquad A(a,b) = I(a,b) + 2, \qquad E(a,b) = I(a,b) + 3,$$
$$N(a) = I(a,1), \qquad X(a,b) = N(A(a,N(b))).$$

They are numerical constructors, not free function symbols in the final polynomial. The eight unconditional axiom-code families are

$$\begin{aligned}
r_1(a,b) &= X(a,E(a,b)),\\
r_2(a,b,c) &= I(A(a,I(b,c)),I(A(a,b),A(a,c))),\\
r_3(a,b,c) &= I(E(a,b),I(M(c,a),M(c,b))),\\
r_4(a,b,c) &= I(A(a,A(b,c)),A(b,A(a,c))),\\
r_5(a,b,c) &= I(E(c,a),I(M(c,b),M(a,b))),\\
r_6(a,b) &= I(X(a,A(a,b)),b),\\
r_G(a,b,c) &= I(E(c,a),I(E(c,b),A(M(a,b),E(a,b)))),\\
r_8(a,b,c,d) &= I(I(I(c,d),a),I(I(a,c),I(b,c))).
\end{aligned}$$

For the guarded bundle define

$$\begin{aligned}
R(b,k,h) &= A(b,I(h,N(A(b,I(h,N(M(b,k))))))),\\
S(a,b,k) &= A(k,I(M(k,a),M(k,b))),\\
C_0(a,b,k,h) &= I(S(a,b,k),R(b,k,h)),\\
C_1(a,b,k,h) &= N(A(a,C_0(a,b,k,h))),\\
C_2(a,b,k,h) &= N(A(a,I(h,C_1(a,b,k,h)))),\\
C(a,b,k,h) &= A(b,I(M(b,k),C_2(a,b,k,h))),\\
V(a,b,c) &= I(M(a,b),A(c,M(a,b))),\\
K(a,b,c) &= A(c,I(C(a,b,c,c),N(R(b,c,c)))),\\
\mathcal{B}(a,b,c) &= N(I(V(a,b,c),K(a,b,c))).
\end{aligned}$$

Its side condition is $c \neq a, c \neq b$, represented by one natural d through

$$(c-a)^2(c-b)^2 = (d+1)^2.$$

For completeness take $d = |(c-a)(c-b)| - 1$.

The twelve padded branches for premises p, q and conclusion v are truth $p = q = v = 0$; the eight axiom branches $p = q = 0, v = r_i$; the guarded bundle $p = q = 0, v = \mathcal{B}(a,b,c)$; modus ponens $q = I(p,v)$; and generalization $q = 0, v = A(a,p)$. The four schema coordinates are shared across the disjunction, but only the selected branch must hold. Squaring each branch's residuals and multiplying the branch sums implements exactly that convention.

The source audit matches the composed constructors to the pinned branch identities. The calibration to ZFC refutability uses the named repository theorems; it is not inferred merely from a resemblance between these integers and logical formulas.

References

[1] Vladimir Reshetnikov and contributors, *ProveIt*, comparison snapshot `a88aefd7ddab4771534bb049c67350e397c91b9a`. https://github.com/VladimirReshetnikov/ProveIt/tree/a88aefd7ddab4771534bb049c67350e397c91b9a.

[2] ProveIt, *Later arrivals and current size reductions*, `Computability/DiophantineConsistency/PROPOSAL_REVIEW_LATEST.md`, at the comparison snapshot.

[3] ProveIt, *Wave11 review and size-reduction decisions*, `Computability/DiophantineConsistency/PROPOSAL_REVIEW_WAVE11.md`, at the comparison snapshot.

[4] ProveIt incoming archive, `ConZFC_Shared_Certificates_2026-10-24.zip`, *Direct proof certificates and smaller Diophantine consistency equations*. The archive filename is preserved as a source identifier. Immediate source files, notices and hashes accompany this contribution.

[5] ProveIt, *A smaller complete source specification for direct proof flow*, `Computability/DiophantineConsistency/candidates/direct-flow-large-carrier/README.md`, at the comparison snapshot; and the inherited `COMMUTATIVE_CERTIFICATES.md`.

[6] Mario Carneiro and Mathlib contributors, *Pell's equation and Matiyasevic's theorem*, `Mathlib/NumberTheory/PellMatiyasevic.lean`, especially `Pell.matiyasevic` and `Pell.eq_pow_of_pell`. https://leanprover-community.github.io/mathlib4_docs/Mathlib/NumberTheory/PellMatiyasevic.html. The archive includes the repository's pinned literal representation interfaces.

[7] ProveIt incoming archive, `ConZFC_13_Integer_Norms_2026-10-24.zip`. The unchanged parent package and its coefficient provenance are included with the thirteen-integer supplement.